# The sign pattern of Sun's trigonometric permanents and a counterexample to his Conjecture 4.6

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**Abstract**

In *Arithmetic properties of some permanents* (arXiv:2108.07723), Zhi-Wei Sun proved that $s_n := (2^{(n-1)/2}/\sqrt{n})\,\mathrm{per}[\sin 2\pi jk/n]_{1\le j,k\le(n-1)/2}$ and $s'_p := (\sqrt{p}/2^{(p-1)/2})\,\mathrm{per}[\csc 2\pi jk/p]_{1\le j,k\le(p-1)/2}$ are integers, and conjectured (Conjecture 4.6) that (i) $n \mid s_n$ for odd composite $n$, and (ii) for odd primes, $s_p < 0 \iff p \equiv 5 \pmod{12}$ and $s'_p < 0 \iff p \equiv 7 \pmod 8$. We refute part (ii) — hence the conjecture as stated — at $p = 29$, the first prime beyond Sun's published table: $s_{29} = 1\,053\,859 > 0$ although $29 \equiv 5 \pmod{12}$, and $s'_{29} = -4\,806\,838\,304 < 0$ although $29 \equiv 5 \pmod 8$. The mod-12 clause fails again at $p = 41$ and 53, the mod-8 clause at $p = 61$. Every value is certified exactly, by at least two of four independently written exact programs, via a cyclotomic identity evaluated in two ways: big-integer arithmetic in $\mathbb{Z}[x]/(x^{4n}-1)$ (the counterexample pair, among others), and finite-field specialization with CRT uniqueness proofs. The pipeline reproduces all nineteen of Sun's published values, and every new prime value satisfies his proven congruences. Part (i) survives all tests (the fifteen odd composites $n \le 65$) and remains open.

## 1 Introduction

In [5], Sun studies permanents of trigonometric matrices and proves, among many results, the following integrality theorem (Theorem 1.6 of [5]). For odd $n > 1$ write $m = (n-1)/2$. Then

$$s_n := \frac{2^m}{\sqrt{n}}\,\mathrm{per}\Big[\sin 2\pi\frac{jk}{n}\Big]_{1\le j,k\le m} \in \mathbb{Z} \qquad \text{and} \qquad s'_p := \frac{\sqrt{p}}{2^m}\,\mathrm{per}\Big[\csc 2\pi\frac{jk}{p}\Big]_{1\le j,k\le m} \in \mathbb{Z}, \tag{1}$$

the first for all odd $n > 1$, the second for all odd primes $p$.[1] Here $\mathrm{per}\,A = \sum_{\tau\in S_m}\prod_{j=1}^m a_{j\tau(j)}$ is the permanent. The same theorem proves the congruences

$$s_p \equiv (-1)^{(p+1)/2} \pmod p \qquad \text{and} \qquad s'_p \equiv 1 \pmod p \tag{2}$$

for every odd prime $p$. In Remark 1.6, Sun reports the values $s_3, \dots, s_{23}$ and $s'_3, \dots, s'_{23}$ (nineteen integers, reproduced in Table 3 below), and in Section 4 he poses, motivated by those data:[2]

**Conjecture 4.6** (Sun [5]). *(i) If $n > 1$ is odd and composite, then $s_n \equiv 0 \pmod n$. (ii) Let $p$ be an odd prime. Then*

$$s_p < 0 \iff p \equiv 5 \pmod{12}, \qquad \textit{and} \qquad s'_p < 0 \iff p \equiv 7 \pmod 8.$$

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[1] In the arXiv TEX source the second definition is typeset "$s'_p$ =:", an evident typo for ":="; the intended reading is unambiguous from the surrounding theorem.

[2] Conjectures in [5] are numbered by section; the statement quoted here is the sixth conjecture of Section 4, i.e. Conjecture 4.6 of the compiled paper (verified against the compiled PDF of arXiv:2108.07723v7). It should not be confused with the identically numbered conjectures of other papers of Sun.

We disprove the conjecture by computing $s_p$ and $s'_p$ exactly for the nine primes $29 \le p \le 61$ and $s_n$ for all fifteen odd composite $n \le 65$. Both clauses of part (ii) already fail at $p = 29$, the first prime beyond the published table (Theorem 2.1). Part (i), by contrast, is *not* refuted: $n \mid s_n$ holds in every tested case (Remark 2.2), and that clause remains an open and, on this evidence, plausible conjecture.

The matrices are dense, with $m$ up to 32, and the values grow to 31 digits, so the computation's trustworthiness deserves comment. Every value reported here was computed by *exact* methods only: an identity in the cyclotomic ring $\mathbb{Z}[\zeta_{4n}]$ (Lemma 3.1) reduces each $s_n$, $s'_p$ to an exactly verifiable integer equation, which was evaluated in two ways — exact big-integer arithmetic in $\mathbb{Z}[x]/(x^{4n} - 1)$ (for the counterexample pair and the small cases), and finite-field specialization at several primes $q \equiv 1 \pmod{4n}$ with Chinese-remainder uniqueness proofs (Section 3). Five independently written programs were used — four exact, plus a high-precision check by Glynn's formula; every value reported here was certified by at least two of the exact programs, with no disagreement observed anywhere, and every new prime value satisfies the proven congruences (2) (Section 4).

Related recent work concerns the *determinant* analogues: Gao and Guo [2] resolve a conjecture of Sun on a trigonometric sec-determinant (the csc analogue having been settled in earlier work of the same authors, cited there). We are not aware of any prior work on the permanent conjecture treated here; as of June 2026 the arXiv record of [5] is unchanged since 2022 (v7), and neither of the sequences $(s_n)$, $(s'_p)$ appears in the OEIS, so Tables 1 and 2 also extend the published data — thirty new exact values — beyond Sun's table.

**Methods and AI-use disclosure.** The counterexamples were found and verified by Demonstrandum, a verification-first multi-agent AI search-and-reasoning pipeline built on Claude language models (Anthropic): agents froze the conjecture statement from the arXiv TEX source of [5], wrote the attack code, and then wrote the verification layers of Section 4 independently of one another and of the attack code. An adversarial audit was performed by a model from a different family (Codex, GPT-5.5, OpenAI), which probed the statement reading and the verification mathematics and independently recomputed the kill pair $(s_{29}, s'_{29})$ and $s_{41}$ at 100–120-digit precision. All accept paths run on exact integer and finite-field arithmetic; floating point appears only in advisory cross-checks and in the generously padded magnitude bounds of one CRT program (Section 3), and every value is also certified by a layer whose bounds are exact-rational or that needs no magnitude bound at all. The text of this note was likewise drafted with the Claude-based pipeline; the author directed the work and takes responsibility for its content. Every number appearing in this note was recomputed or re-verified from the certificate data during its preparation (June 2026); see Section 4.

## 2 The counterexamples

**Theorem 2.1.** *Conjecture 4.6 of [5] is false. Both clauses of part (ii) fail at $p = 29$, where $29 \equiv 5 \pmod{12}$ and $29 \equiv 5 \pmod 8$:*

$$s_{29} \;=\; 1\,053\,859 \;>\; 0, \qquad\qquad s'_{29} \;=\; -4\,806\,838\,304 \;<\; 0,$$

*so the implication $p \equiv 5 \pmod{12} \Rightarrow s_p < 0$ fails, and so does the implication $s'_p < 0 \Rightarrow p \equiv 7 \pmod 8$. Moreover the first clause fails at every prime $p \equiv 5 \pmod{12}$ in the computed range ($p = 29, 41, 53$, with $s_{41} = 5\,574\,476\,521$ and $s_{53} = 1\,540\,679\,755\,916\,971$, both positive), and the second clause fails again at $p = 61 \equiv 5 \pmod 8$, where*

$$s'_{61} \;=\; -2\,904\,784\,276\,786\,469\,053\,142\,518\,062\,479 \;<\; 0.$$

*Proof.* A finite exact computation. The values of $s_p$ and $s'_p$ for the nine primes $29 \le p \le 61$ are listed in Table 1; each is certified, via the cyclotomic method of Section 3, by at least two independently written exact programs with disjoint prime sets (the kill pair additionally by the exact big-integer implementation and a third CRT program), validated as described in Section 4. The residues are elementary. Since both violations are sign statements about nonzero integers, no boundary or strictness issue arises; note that $s'_p = 0$ is in any case excluded by the proven congruence (2). □

| $p$ | $p$ mod 12 | $p$ mod 8 | $s_p$ | $s'_p$ | violates |
|---|---|---|---|---|---|
| 29 | 5 | 5 | 1 053 859 | −4 806 838 304 | both clauses |
| 31 | 7 | 7 | 10 542 977 | −1 518 806 869 720 | — |
| 37 | 1 | 5 | 1 519 663 259 | 43 041 655 439 377 | — |
| 41 | 5 | 1 | 5 574 476 521 | 88 188 655 594 502 880 | mod-12 clause |
| 43 | 7 | 3 | 453 435 884 081 | 7 817 331 967 711 147 274 | — |
| 47 | 11 | 7 | 4 570 760 060 257 | −208 210 243 110 377 949 730 | — |
| 53 | 5 | 5 | 1 540 679 755 916 971 | 1 364 915 376 262 405 923 148 317 | mod-12 clause |
| 59 | 11 | 3 | 493 044 351 638 203 633 | 7 313 054 784 852 201 235 037 895 089 487 | — |
| 61 | 1 | 5 | 3 864 901 746 617 921 299 | −2 904 784 276 786 469 053 142 518 062 479 | mod-8 clause |

Table 1: Exact values of $s_p$ and $s'_p$ for the nine primes beyond Sun's published table. The mod-12 clause of Conjecture 4.6(ii) is violated iff ($p \equiv 5 \pmod{12}$ and $s_p > 0$) or ($p \not\equiv 5$ and $s_p < 0$); the mod-8 clause iff ($p \equiv 7 \pmod 8$ and $s'_p > 0$) or ($p \not\equiv 7$ and $s'_p < 0$). Every value satisfies the proven congruences (2).

*Remark* 2.2 (Part (i) is not refuted). The divisibility clause was tested exactly for *all* fifteen odd composite $n \le 65$, including the prime squares 9, 25, 49 and the values $25, 35, 49, 55, 65$ coprime to 3; Table 2 lists $s_n$ and the integer quotient $s_n/n$. The divisibility $n \mid s_n$ holds in every case. Part (i) of Conjecture 4.6 therefore survives this attack and remains open; our refutation of the conjecture (a conjunction) rests entirely on part (ii).

| $n$ | $s_n$ | $s_n/n$ | $n$ | $s_n$ | $s_n/n$ |
|---|---|---|---|---|---|
| 9 | 9 | 1 | 45 | 696 741 294 375 | 15 483 139 875 |
| 15 | 45 | 3 | 49 | 3 078 195 713 761 | 62 820 320 689 |
| 21 | 2 835 | 135 | 51 | 113 109 498 706 113 | 2 217 833 307 963 |
| 25 | 63 125 | 2 525 | 55 | 9 553 828 212 328 125 | 173 705 967 496 875 |
| 27 | 59 049 | 2 187 | 57 | 89 237 136 634 668 369 | 1 565 563 800 608 217 |
| 33 | −1 643 895 | −49 815 | 63 | 14 357 464 732 984 700 313 | 227 896 265 602 931 751 |
| 35 | 334 186 125 | 9 548 175 | 65 | 23 595 726 326 056 828 125 | 363 011 174 247 028 125 |
| 39 | 9 154 298 673 | 234 725 607 | | | |

Table 2: The part-(i) test set: exact $s_n$ for all odd composite $n \le 65$. The quotient $s_n/n$ is an integer in every case, so part (i) of Conjecture 4.6 is *not* refuted. (The values for $n = 9, 15, 21$ agree with Sun's table.)

*Remark* 2.3 (Why the pattern broke). Each sign law of part (ii) was fitted to eight primes containing only two negatives each ($s_p < 0$ at $p = 5, 17$; $s'_p < 0$ at $p = 7, 23$). In the extended range the first law collapses completely: $s_p > 0$ for *every* prime $29 \le p \le 61$, so the only known negative values of $s_p$ remain $p = 5$ and 17, and the law fails at every tested $p \equiv 5 \pmod{12}$ beyond the published table. Negative values of $s'_p$ do persist — at $p = 31$ and 47, both $\equiv 7 \pmod 8$, the second law happens to conform — but they also appear at $p = 29$ and 61, both $\equiv 5 \pmod 8$. Galois equivariance does not separate the permanent from its determinant cousins here: the automorphism $\sigma_a \colon \zeta_p \mapsto \zeta_p^a$ of $\mathbb{Q}(\zeta_p)$ multiplies per $M$ (the matrix of Lemma 3.1 with $n = p$; its entries $\zeta_p^{jk} - \zeta_p^{-jk}$ lie in $\mathbb{Z}[\zeta_p]$) by the Legendre symbol $(\frac{a}{p})$ — the computation at the heart of Sun's integrality proof — so the permanent, too, is a Galois eigenvector, and that symmetry constrains the sign of neither quantity. What the determinant analogues studied by Sun and in [2] additionally possess is a character factorization: their determinants factor over Dirichlet characters into character sums — in [2], into special values of Dirichlet $L$-functions — and structure of that kind can force residue-class behaviour. No analogous factorization is known for the permanent, and we know of no mechanism that could enforce a residue-class sign rule for $s_p$ or $s'_p$. We do not propose a replacement law: the observed signs describe the computed range $p \le 61$ only.

## 3 The cyclotomic verification method

The permanents in (1) are sums of $m!$ products of irrational reals, the results are integers as large as $10^{30}$, and a sign is precisely what is at stake; the following identity removes all analysis from the problem.

For odd $n > 1$ let $\zeta = e^{2\pi i/(4n)}$, a primitive $4n$-th root of unity, so that $i = \zeta^n$ and $e^{2\pi i\, jk/n} = \zeta^{4jk}$, and let

$$G \;=\; \sum_{a=0}^{n-1} \zeta^{4a^2} \;\in\; \mathbb{Z}[\zeta]$$

be the quadratic Gauss sum. By Gauss's theorem (see, e.g., [1, Thm. 1.5.2]), $G = \sqrt{n}$ if $n \equiv 1 \pmod 4$ and $G = i\sqrt{n}$ if $n \equiv 3 \pmod 4$. Set $\widetilde{G} = G$ in the first case and $\widetilde{G} = \zeta^{-n} G = -iG$ in the second, so that $\widetilde{G} = \sqrt{n}$ for every odd $n$; in particular $\sqrt{n} \in \mathbb{Z}[\zeta]$.

**Lemma 3.1.** *Let $n > 1$ be odd, $m = (n-1)/2$, and $M_{jk} = \zeta^{4jk} - \zeta^{-4jk}$ for $1 \le j, k \le m$. Then*

$$\operatorname{per} M \;=\; s_n\, i^m \sqrt{n} \qquad \text{in } \mathbb{Z}[\zeta]. \tag{3}$$

*If moreover $n = p$ is prime, the matrix $U$ with entries $U_{jk} = p\,(\zeta^{4jk} - \zeta^{-4jk})^{-1}$ has entries in $\mathbb{Z}[\zeta]$, and*

$$\operatorname{per} U \;=\; s'_p\, p^{m-1} \sqrt{p}\, i^{-m} \qquad \text{in } \mathbb{Z}[\zeta]. \tag{4}$$

*Proof.* From $\sin\theta = (e^{i\theta} - e^{-i\theta})/(2i)$ we get $\sin(2\pi jk/n) = M_{jk}/(2i)$, and multilinearity of the permanent in the rows gives $\operatorname{per}[\sin 2\pi jk/n] = (2i)^{-m} \operatorname{per} M$; substituting into (1) yields (3). For (4), $\csc(2\pi jk/p) = 2i\, M_{jk}^{-1}$, so $\operatorname{per}[\csc 2\pi jk/p] = (2i)^m p^{-m} \operatorname{per} U$, and (1) gives $\operatorname{per} U = s'_p\, p^m i^{-m}/\sqrt{p} = s'_p\, p^{m-1} \sqrt{p}\, i^{-m}$. The entries of $U$ are algebraic integers: $\zeta^{4jk} - \zeta^{-4jk} = \zeta^{-4jk}(\omega^{jk} - 1)$ with $\omega = \zeta^8$ a primitive $p$-th root of unity and $p \nmid jk$, and $\omega^{jk} - 1$ divides $p = \prod_{t=1}^{p-1}(1 - \omega^t)$ in $\mathbb{Z}[\omega]$. $\square$

Both sides of (3) and (4) lie in $\mathbb{Z}[\zeta] \cong \mathbb{Z}[x]/(\Phi_{4n}(x))$, where $\Phi_{4n}$ is the $4n$-th cyclotomic polynomial, and the right-hand sides are nonzero integer multiples of a fixed ring element; hence each identity determines the integer $s_n$ (resp. $s'_p$) uniquely and reduces its computation to finite integer arithmetic. We used two independent implementations.

**Implementation A: exact arithmetic in $\mathbb{Z}[x]/(x^{4n}-1)$.** Represent elements of $\mathbb{Z}[\zeta]$ by integer polynomials modulo $x^{4n} - 1$, with $x$ standing for $\zeta$. The permanent of $M$ (or $U$) is computed by Ryser's inclusion–exclusion formula [4] over this ring, a Gray-code walk over the $2^m$ column subsets; for speed, ring elements are packed into pairs of nonnegative big integers by Kronecker substitution $x \mapsto 2^K$, with $K$ chosen from a rigorous a-priori $L^1$ coefficient bound so that no base-$2^K$ digit can overflow, and cyclic reduction modulo $x^{4n} - 1$ becomes arithmetic modulo $2^{4nK} - 1$ (every decode is re-encoded and asserted equal as an integrity check). The program then *solves* for the unique integer $t$ with $P \equiv t\,B \pmod{\Phi_{4n}(x)}$ over $\mathbb{Z}$, where $P$ is the computed permanent and $B = x^{nm}\, \widetilde{G}(x)$ (resp. the right side of (4)); $\Phi_{4n}$ is computed by exact division of $x^{4n} - 1$ by lower cyclotomic polynomials. For the csc case the entries $U_{jk}$ are obtained by the extended Euclidean algorithm over $\mathbb{Q}[x]$ modulo $\Phi_{4p}$ and verified by exact back-multiplication $U_{jk} \cdot (x^{4jk} - x^{-4jk}) \equiv p \pmod{\Phi_{4p}}$ over $\mathbb{Z}$. No floating point and no finite fields are used anywhere. On a desktop machine this implementation delivers $s_{29}$ in about four and a half minutes and $s'_{29}$ in about seven.

**Implementation B: finite-field specialization and CRT uniqueness.** For a prime $q \equiv 1 \pmod{4n}$, the field $\mathbb{F}_q$ contains elements $w$ of exact multiplicative order $4n$; any such $w$ is a root of $\Phi_{4n}$ modulo $q$, so reduction modulo $q$ and evaluation $x \mapsto w$ is a ring homomorphism $\mathbb{Z}[x]/(\Phi_{4n}(x)) \to \mathbb{F}_q$. The images of (3)–(4) are then identities in $\mathbb{F}_q$ in which the image $g$ of $\widetilde{G}$ is computed from the defining sum $\sum_a w^{4a^2}$, multiplied by $w^{-n}$ when $n \equiv 3 \pmod 4$ (so no square-root sign ambiguity can arise; the implementations assert $g^2 \equiv n \pmod q$), the permanent of the specialized matrix is computed modulo $q$, and $s_n \bmod q$ is read off. Repeating this for several random primes $q$ and combining by the Chinese remainder theorem with $\prod q > 4B$, where $B$ is a bound on the magnitude of the target integer ($|s_n| \le 2^m m!$ suffices for the sin case; for the csc case an explicit bound follows from the chord inequality $|\sin \pi x| \ge 2 \operatorname{dist}(x, \mathbb{Z})$), determines the integer *uniquely* — a proof, not a probabilistic check, provided $B$ itself is rigorous. Three independently written programs of this type were run, with disjoint random prime sets and different permanent kernels: a Gray-code Ryser in C with Montgomery arithmetic (59-bit primes; generously padded floating-point bounds, with every value additionally confirmed at a held-out prime not used in the reconstruction), a plain 128-bit-integer Ryser in C (fresh primes from $[2^{60}, 2^{61}]$; bounds evaluated in exact rational arithmetic), and a pure-Python subset-dynamic-programming permanent (57-bit primes; exact rational bounds and a held-out prime per value), the last using the recursion $f(S) = \sum_{j \in S} A_{|S|,j}\, f(S \setminus \{j\})$ rather than Ryser's formula. The $[2^{60}, 2^{61})$ run certified 32 values — the kill pair, every $s_n$ for odd $31 \le n \le 65$, every $s'_p$ for $31 \le p \le 61$, and four values from Sun's table — in about 22 minutes of wall-clock time.

Implementations A and B share no code; A was written by an agent with no access to the discovery code. In addition to the exact layers, two floating-point recomputations were performed as advisory cross-checks: Glynn's formula [3] — a third permanent algorithm — in 60-digit arithmetic, and, during the adversarial audit, independent 100–120-digit recomputations of $s_{29}$, $s'_{29}$ and $s_{41}$ by both Glynn's and Ryser's formulas directly from the real entries (residuals below $10^{-86}$ in all cases).

# 4 Verification and calibration

**Calibration on Sun's table.** Before computing anything new, the pipeline was calibrated against the nineteen values published in Remark 1.6 of [5] (Table 3): three of the five programs — implementation A, the first CRT program, and the subset-DP program — reproduced all nineteen exactly, and the remaining two were first validated on subsets containing negative entries (the clean-room CRT program on $s_{17}$, $s_{23}$, $s'_{17}$, $s'_{23}$; the Glynn check on its anchors). The four negative entries ($s_5$, $s_{17}$, $s'_7$, $s'_{23}$) pin the sign conventions end-to-end: a global sign error in any layer — in the Gauss sum, in the power of $i$, or in the permanent kernel — would flip at least one of them.

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $n$ | 3 | 5 | 7 | 9 | 11 | 13 | 15 | 17 | 19 | 21 | 23 |
| $s_n$ | 1 | −1 | 1 | 9 | 1 | 51 | 45 | −239 | 913 | 2 835 | 12 145 |
| $p$ | 3 | 5 | 7 | 11 | 13 | 17 | 19 | 23 | | | |
| $s'_p$ | 1 | 1 | −6 | 111 | 261 | 6 784 | 245 101 | −7 094 142 | | | |

Table 3: Calibration: the nineteen values published by Sun [5, Remark 1.6], all reproduced exactly — by three of the five programs in full, by the other two on validation subsets — before any new value was computed.

**Cross-checks on the new values.** The exact programs of Section 3 agree wherever their coverage overlaps, and every entry of Tables 1 and 2 is certified by at least two of them.[3] Beyond mutual agreement: every $s_p$, $s'_p$ for $29 \le p \le 61$ satisfies the congruences (2), which are *proven* in [5] — the only theorem available to test the new values against (for the kill pair: $s_{29} \equiv 28 \equiv (-1)^{15}$ and $s'_{29} \equiv 1 \pmod{29}$); the first and third CRT programs confirmed each of their values — between them, every value in this note — at a held-out prime; and the adversarial audit (three rounds, Codex/GPT-5.5) probed the statement provenance (permanent vs. determinant, the half-range index set, the conjecture numbering) and recomputed the headline values independently at high precision.

**Reproducibility.** All computations were re-run while preparing this note (June 2026). A reader can reproduce the kill in under two minutes: a one-page floating-point script (Ryser's formula in IEEE doubles) reproduces $s_{17}$, $s'_7$, $s'_{23}$ and the kill pair to the nearest integer in about a second, and the pure-Python exact program (implementation B, subset-DP kernel) re-derives all nineteen calibration values, the kill pair, and $s_{41}$, with full CRT uniqueness proofs, in under a minute. While drafting this note we additionally re-verified the congruences (2) and the residues and quotients of Tables 1–2, and recomputed, by a freshly written floating-point Gray-code Ryser, every table entry with $m \le 18$ (all nineteen calibration values; $s_p$, $s'_p$ for $p = 29, 31, 37$; $s_n$ for $n = 25, 27, 33, 35$).

**Data availability.** The certificate bundle (all five programs, their logs, machine-readable certificates with the 33 exact integers of Tables 1–2 and their provenance, and the frozen arXiv source of [5]) will be made available at `https://github.com/demonstrandum-research/[REPO]`.

[3] Coverage detail: implementation A computed the kill pair and the small composites exactly; the first CRT program computed every value; the clean-room CRT program certified the 32 values listed above; the subset-DP program covers the kill pair and $s_{41}$. Thus the headline values are certified by three or four exact programs plus the floating-point layers. The supplementary values $s_{53}$ and $s'_{61}$ rest on the two independent CRT programs alone and were not part of the 100-digit audit recomputation.

# References


[1] B. C. Berndt, R. J. Evans, and K. S. Williams, *Gauss and Jacobi Sums*, Canadian Mathematical Society Series of Monographs and Advanced Texts **21**, Wiley, New York, 1998.

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